tugturn: A Markerless 3D Pipeline for Instrumented Timed Up and Go Phase Segmentation, Gait Events, and Stability Metrics

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Abstract

Instrumented Timed Up and Go (TUG) analysis can support clinical and research decision-making, but robust and reproducible markerless pipelines are still limited. We present *tugturn*, a Python-based workflow for 3D markerless TUG processing that combines phase segmentation, gait-event detection, spatiotemporal metrics, intersegmental coordination, and dynamic stability analysis. The pipeline uses spatial thresholds to segment each trial into stand, forward gait, turning, backward gait, and sit phases, and applies a relative-distance strategy to detect heel-strike and toe-off events within valid gait windows. In addition to conventional kinematics, *tugturn* provides Vector Coding outputs and Extrapolated Center of Mass (XCoM)-based metrics. The software is configured through TOML files and produces reproducible artifacts, including HTML reports, CSV tables, and quality-assurance visual outputs. A complete runnable example is provided with test data and command-line instructions. This manuscript describes the implementation, outputs, and reproducibility workflow of *tugturn* as a focused software contribution for markerless biomechanical TUG analysis.

Keywords Timed Up and Go · markerless 3D analysis · gait event detection · Vector Coding · XCoM · biomechanics software

1 Introduction

Timed Up and Go (TUG) is widely used to evaluate mobility and functional performance in clinical populations because it integrates postural transition, walking, turning, and sitting in a single task [Podsiadlo and Richardson, 1991]. Traditional motion-capture workflows provide high-fidelity biomechanical information, but they often depend on specialized infrastructure and time-consuming processing pipelines. Markerless approaches based on computer vision have expanded access to movement analysis and enabled broader deployment in rehabilitation and applied biomechanics [Lugaresi et al., 2019, Nath et al., 2019].

Despite this progress, many markerless pipelines still emphasize either pose extraction or generic gait summaries and do not provide a structured TUG-specific workflow with robust phase-aware event handling. In TUG, meaningful interpretation depends on correctly isolating movement phases (e.g., forward gait versus return gait) and on reducing false event detections outside valid locomotor windows. This issue is particularly relevant in heterogeneous clinical recordings, where pauses, partial turns, and variable trajectories are common.

The *tugturn* module was designed to address this gap by integrating: (i) spatially constrained phase segmentation, (ii) gait-event detection based on relative-distance principles [Zeni et al., 2008], (iii) phase-specific spatiotemporal metrics, (iv) Vector Coding for coordination analysis, and (v) Extrapolated Center of Mass (XCoM) indicators for dynamic

stability assessment. The goal of this paper is to document the software architecture and reproducible processing flow of *tugturn* as a practical tool for markerless 3D TUG analysis.

2 Methods and Implementation

2.1 Input Data and Processing Model

tugturn processes MediaPipe-based 3D CSV files generated from markerless pose estimation pipelines [Lugaresi et al., 2019]. Execution supports both single-file and batch-directory modes through a command-line interface. Subject and trial parameters can be supplied in TOML files, allowing standardized runs across datasets.

The core implementation is organized around a `TUGAnalyzer` class in `tugturn.py`, which computes kinematic variables, segment boundaries, gait events, and report-ready outputs from a unified data structure.

2.2 Phase Segmentation

Each trial is segmented into:

- `stand`
- `gait_forward`
- `turn180`
- `gait_back`
- `sit`

Segmentation combines temporal logic and spatial thresholds along the anteroposterior axis, with configurable TOML parameters:

- `y_chair`
- `y_turn`
- `y_chair_tol`

This design improves robustness when trajectories include pauses, short hesitations, or non-ideal paths.

2.3 Gait Events and Kinematic Metrics

Heel-strike (HS) and toe-off (TO) events are estimated using a relative-distance strategy inspired by the method described by Zeni et al. [2008]. Events are then filtered by expected phase windows to reduce fictitious detections outside valid gait segments.

The software computes:

- joint angles (e.g., knee and ankle, right/left),
- trunk inclination,
- trunk and foot kinematics in the sagittal YZ plane (position, velocity, acceleration),
- global and phase-specific spatiotemporal parameters.

To reduce sign ambiguity in sagittal derivatives, *tugturn* also reports 2D resultant kinematic magnitude:

$$|V_{yz}| = \sqrt{V_y^2 + V_z^2}, \qquad |A_{yz}| = \sqrt{A_y^2 + A_z^2}.$$

2.4 Coordination and Stability Metrics

Vector Coding. The pipeline computes coupling-angle series for trunk–pelvis and limb coordination analyses. Coupling angles are normalized to $[0, 360)$ and classified into canonical coordination patterns (e.g., in-phase and anti-phase classes), with percentages and variability summaries.

XCoM. Dynamic stability is estimated from Extrapolated Center of Mass:

$$XCoM = CoM + \frac{CoM_{vel}}{\omega_0}, \qquad \omega_0 = \sqrt{\frac{g}{l}},$$

where l is approximated by mean vertical CoM. Phase-level XCoM deviations are exported as stability descriptors for forward and backward gait.

2.5 Algorithmic Workflow (Pseudocode)

Input: 3D MediaPipe CSV file, optional TOML metadata/config, optional output path.
Output: HTML reports, JSON report, and CSV databases (participants, results, steps, kinematics, vector coding).

1. Load CSV and TOML metadata; parse FPS and spatial thresholds.
2. Initialize `TUGAnalyzer`, compute CoM, and extract kinematics.
3. Detect HS/TO gait events and segment phases (`stand`, `gait_forward`, `turn180`, `gait_back`, `sit`).
4. Filter events outside valid gait windows and apply spatial validity checks.
5. Compute spatiotemporal metrics and build step-level time series.
6. Deduplicate simultaneous steps and compute phase-level QA artifacts (GIFs).
7. Compute Vector Coding summaries for turn, stand, forward gait, and backward gait.
8. Assemble unified report object (metadata, phases, metrics, steps, vector coding).
9. Export `*_tug_data.json`, `*_bd_participants.csv`, `*_bd_results.csv`, `*_bd_steps.csv`, and `*_bd_kinematics.csv`.
10. Generate static and interactive HTML reports (`*_tug_report.html` and `*_tug_report_interactive.html`).

3 Reproducibility and Usage

3.1 Command-Line Execution

Example runs:

```
# Single file with explicit output directory
uv run vaila/tugturn.py -i tests/tugturn/s26_m1_t1.csv \
  -c tests/tugturn/s26_m1_t1.toml -o tests/tugturn/results/ -y 0.15

# Single file without -o (auto output directory)
uv run vaila/tugturn.py -i tests/tugturn/s26_m1_t1.csv \
  -c tests/tugturn/s26_m1_t1.toml

# Batch processing by directory
uv run vaila/tugturn.py -i tests/tugturn/ \
  -c tests/tugturn/s26_m1_t1.toml
```

3.2 Expected Artifacts

Typical outputs per trial include:

- `*_tug_report.html`
- `*_tug_report_interactive.html`
- `*_bd_results.csv`
- `*_bd_steps.csv`
- `*_bd_kinematics.csv`
- Vector Coding metric and time-series CSV files
- phase-level QA GIF files

4 Results-Oriented Software Deliverables

Rather than presenting a new clinical trial in this manuscript, we report software deliverables and processing scope demonstrated by the included test workflow. The pipeline provides:

- deterministic phase segmentation with explicit spatial constraints,
- phase-aware HS/TO event extraction,
- kinematic and spatiotemporal summaries aligned with TUG sub-tasks,
- coordination and stability metrics in machine-readable tables,
- browser-ready reports for rapid quality control and interpretation.

Table 1 summarizes the core output families.

Table 1: Core output families generated by *tugturn*.

Output family	**Purpose**
HTML reports	Human-readable trial overview and plots
`bd_results` CSV	Global and phase metrics for analysis
`bd_steps` CSV	Step-level event and spatiotemporal values
`bd_kinematics` CSV	Time-series kinematic variables
Vector Coding CSVs	Coupling-angle metrics and variability summaries
Phase GIFs	Segmentation quality assurance

4.1 Example Run Output (HTML Reports)

To demonstrate a real execution artifact in this manuscript, we include the reference run `s26_m1_t1` generated in: `tugturn/results/`. The corresponding HTML outputs are:

- `tugturn/results/s26_m1_t1_tug_report.html`
- `tugturn/results/s26_m1_t1_tug_report_interactive.html`

From the same run, Table 2 summarizes key values exported by `s26_m1_t1_tug_data.json` and `s26_m1_t1_bd_results.csv`.

Table 2: Selected metrics from the example run `s26_m1_t1`.

Metric	**Value**
Turn direction	Right
Total TUG time	26.93 s
Stand phase duration	1.48 s
Forward gait duration	5.81 s
Turn duration	2.79 s
Backward gait duration	5.76 s
Sit phase duration	3.85 s
Cadence	67.91 steps/min
Velocity	0.347 m/s
Steps (forward/backward)	12 / 11
XCoM deviation (forward/backward)	0.027 / 0.041 m

Because LaTeX does not natively render HTML pages as inline figures, the HTML reports are treated as supplementary digital artifacts in this version of the manuscript. They preserve interactive visual inspection while the paper reports the quantitative summary.

5 Discussion

tugturn contributes a targeted software layer for TUG-specific markerless analysis, emphasizing phase-aware processing rather than generic full-trial summaries. This focus is important for protocols where turning, sit-to-stand transitions, and return gait carry distinct clinical meaning.

From a software-engineering perspective, the combination of CLI execution, TOML configuration, and structured outputs facilitates reproducibility, batch scaling, and downstream statistical workflows. The report stack (HTML plus CSV) supports both exploratory interpretation and scripted analysis.

Current limitations include dependence on upstream pose-estimation quality, protocol-specific threshold tuning, and the absence of benchmarked agreement against instrumented laboratory gold standards in this manuscript. Future work should include cross-dataset validation, uncertainty quantification, and multicenter reproducibility studies.

6 Conclusion

This work presents *tugturn* as a reproducible and practical software pipeline for markerless 3D TUG analysis. By combining spatial phase segmentation, phase-filtered gait events, Vector Coding, and XCoM metrics in a single workflow, *tugturn* enables consistent extraction of clinically relevant biomechanical descriptors from routine video-derived data. The provided test workflow and command-line interface support transparent reuse and adaptation in biomechanics and rehabilitation research.

Availability

The software, test files, and execution examples used in this manuscript are available in the project repository. The `tugturn` example dataset and TOML configuration support direct reproducibility of the documented command-line runs.

A Example Run Reports in HTML

The complete result reports for the reference run (`s26_m1_t1`) are provided as HTML artifacts in `tugturn/results/`. These files are the primary run-level result outputs generated by the pipeline:

- `tugturn/results/s26_m1_t1_tug_report.html`
- `tugturn/results/s26_m1_t1_tug_report_interactive.html`

Additional machine-readable outputs from the same run are:

- `s26_m1_t1_tug_data.json`
- `s26_m1_t1_bd_results.csv`
- `s26_m1_t1_bd_steps.csv`
- `s26_m1_t1_bd_kinematics.csv`
- `s26_m1_t1_bd_vector_coding.csv`
- `s26_m1_t1_bd_participants.csv`

In this manuscript, quantitative summaries are reported in the main Results section, while the HTML reports are retained in the appendix as full visual and interactive evidence of the execution output.

The full PDF report exported from this run is included below as appendix pages, preserving the complete clinical output view used for disability-focused cohorts (including Parkinson's disease): `images/TUG Report_ s26_m1_t1.pdf`.

vailá - TUG Analysis Report

Subject: s26_m1_t1

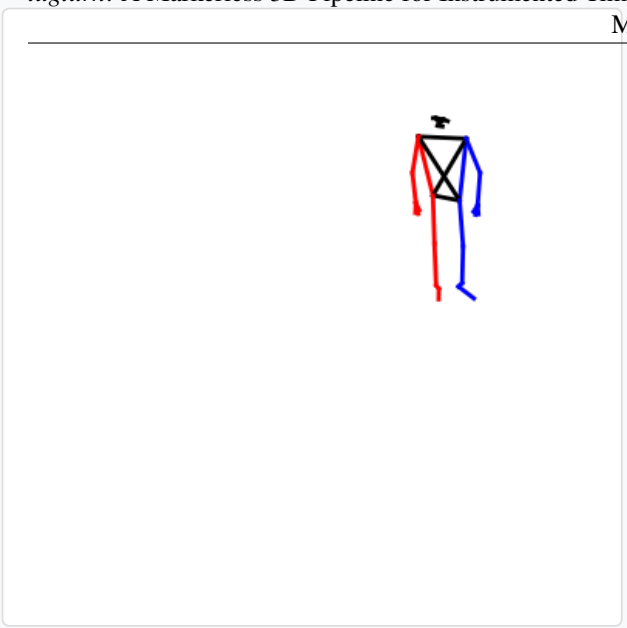
Phase Recordings

Stand

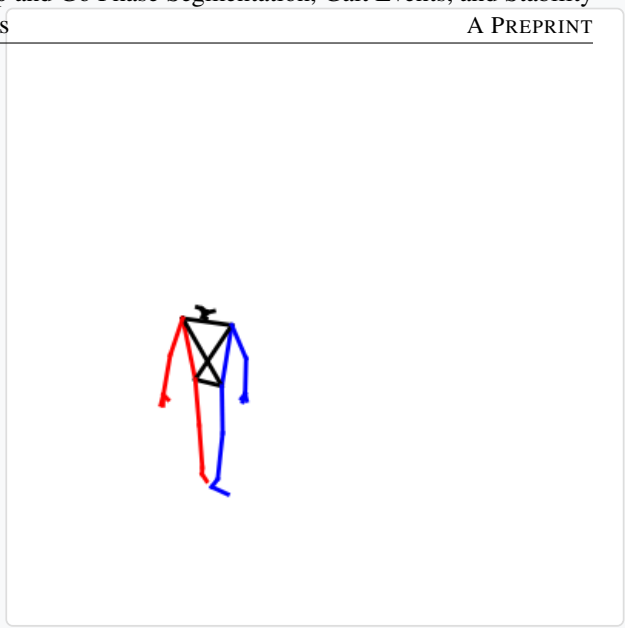
Gait Forward

Stop 5s

Turn180

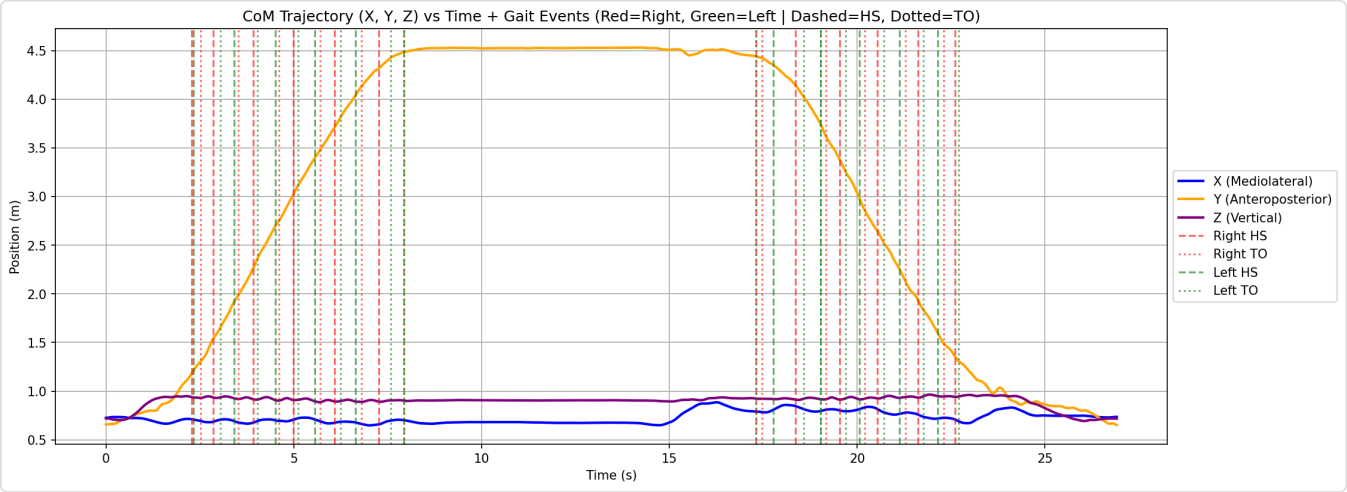


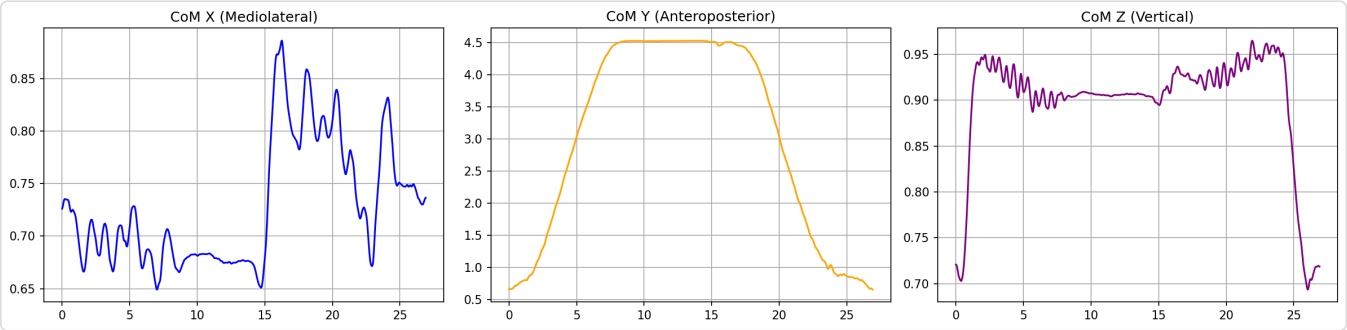
Gait Back

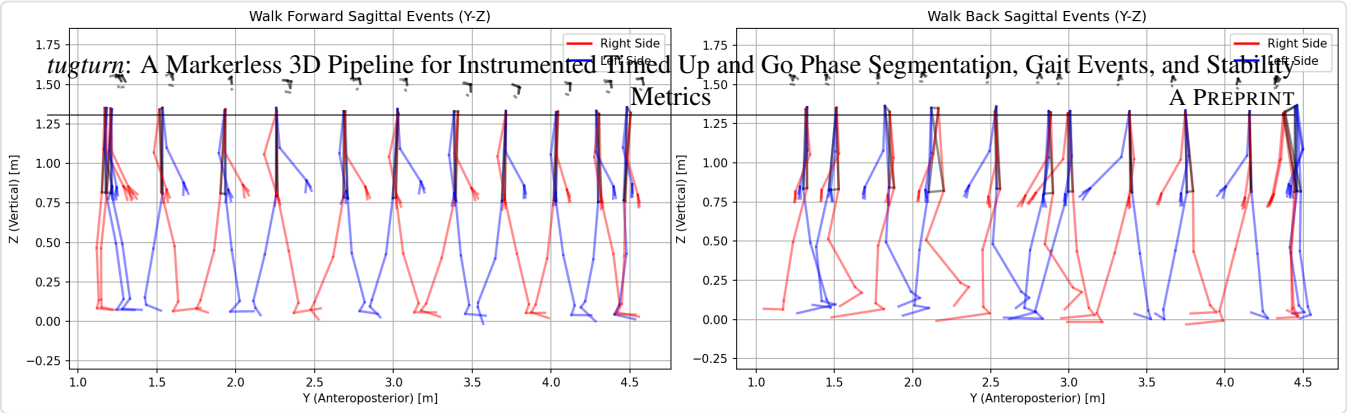


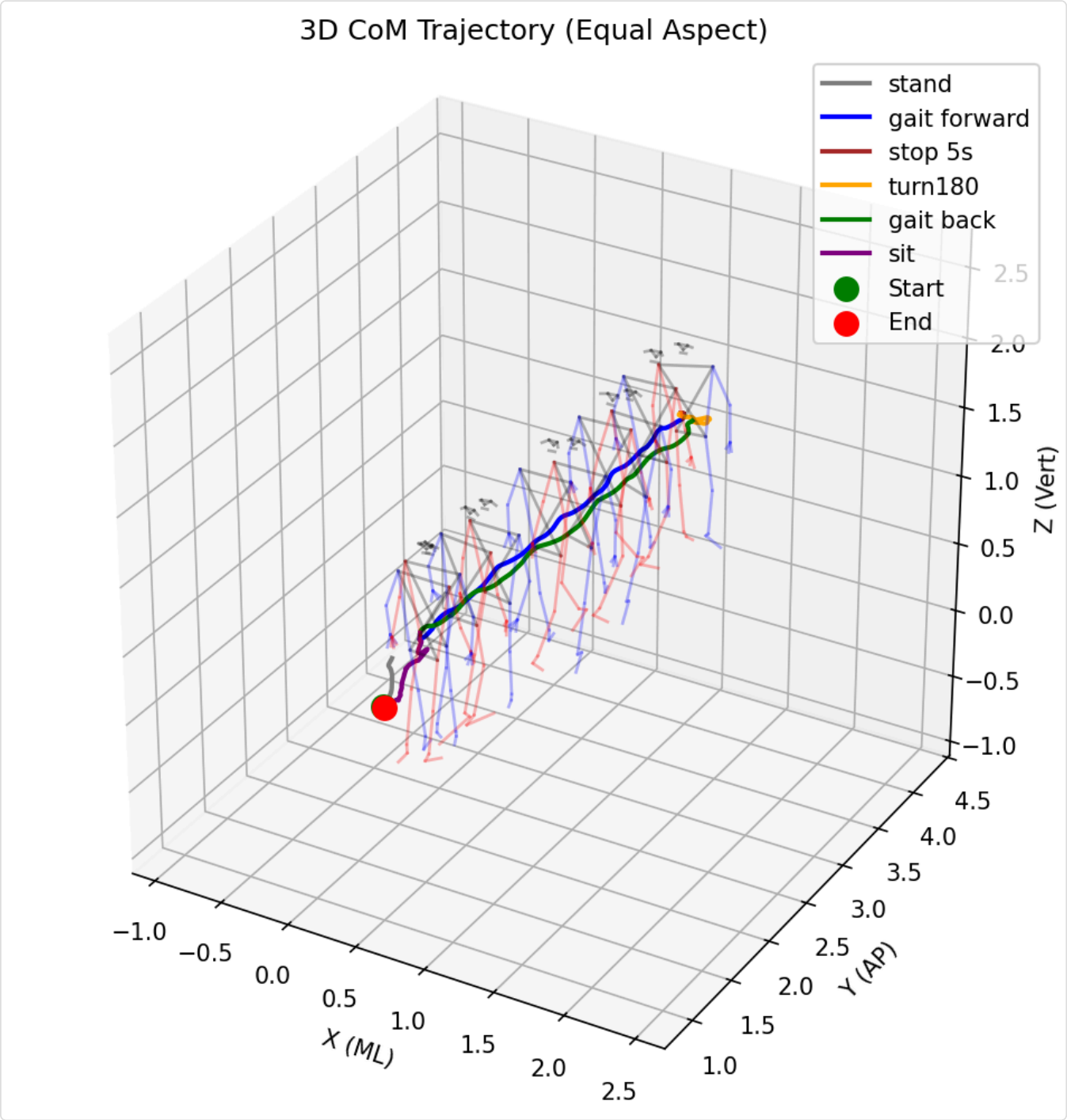
Sit

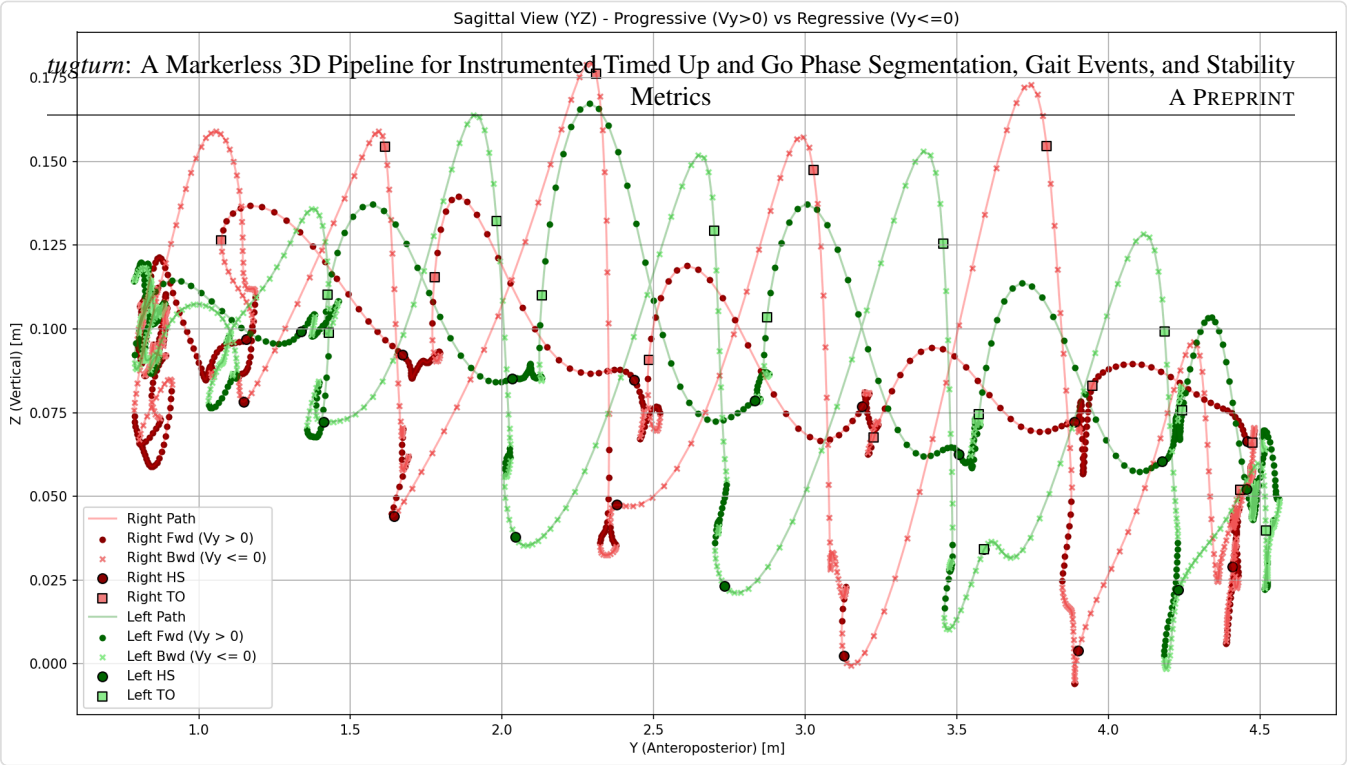
Visual Charts

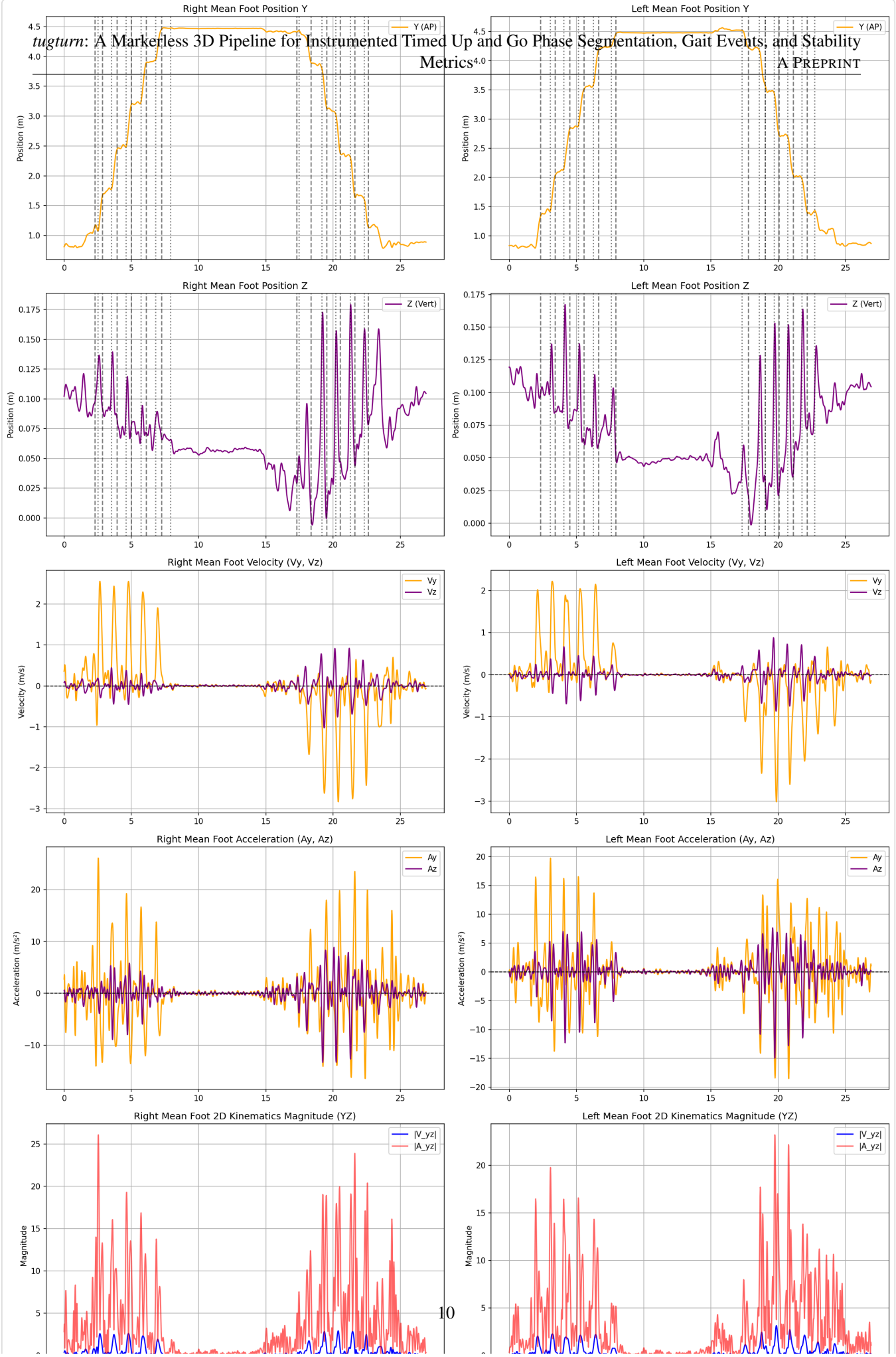


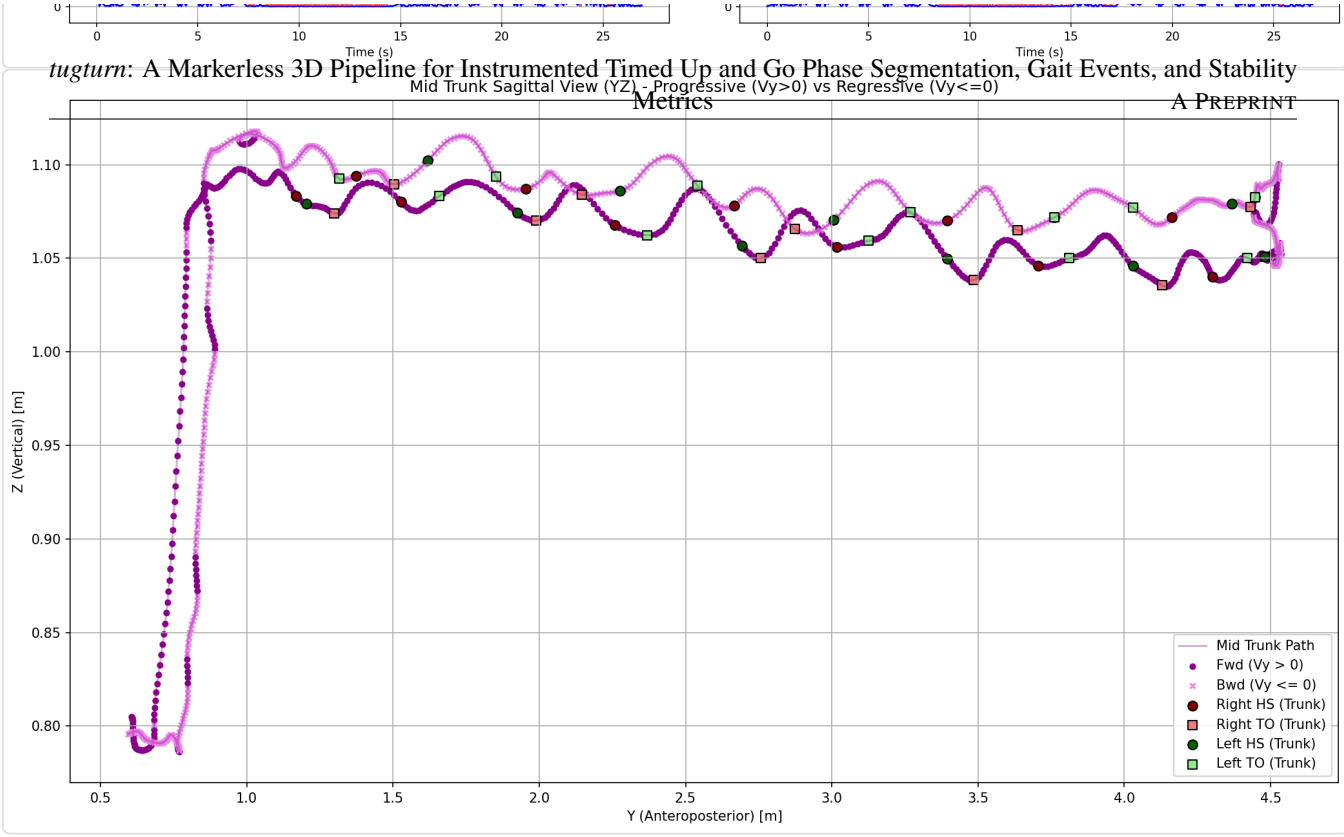


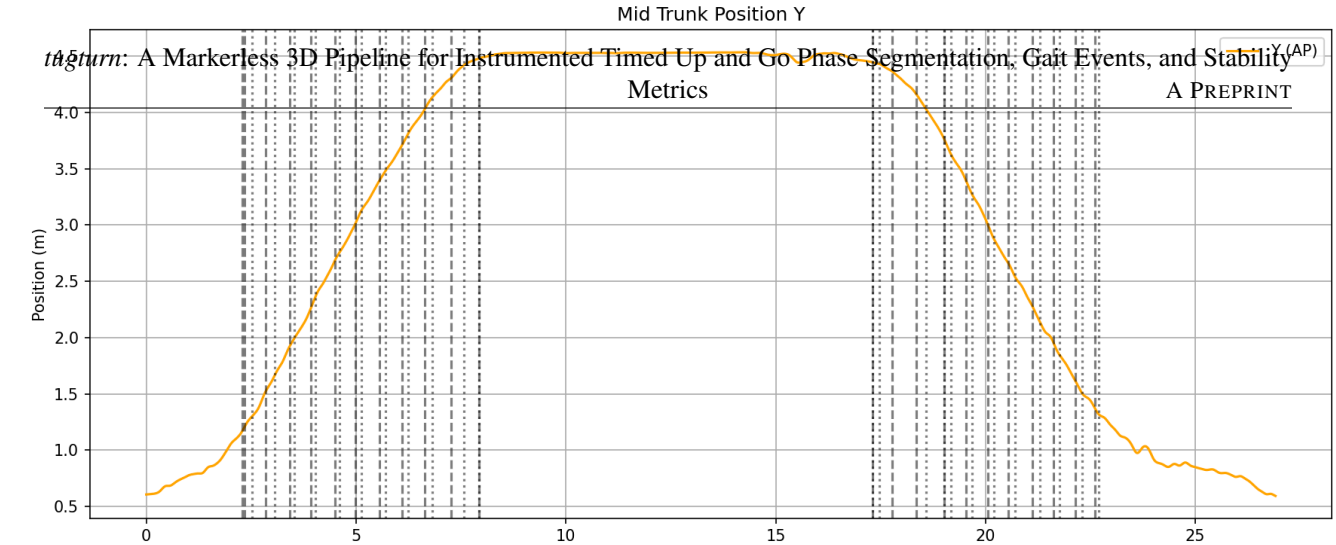

Walk Forward Sagittal Events (Y-Z)
Walk Back Sagittal Events (Y-Z)
Right Side
Left Side
Z (Vertical) [m]
Y (Anteroposterior) [m]

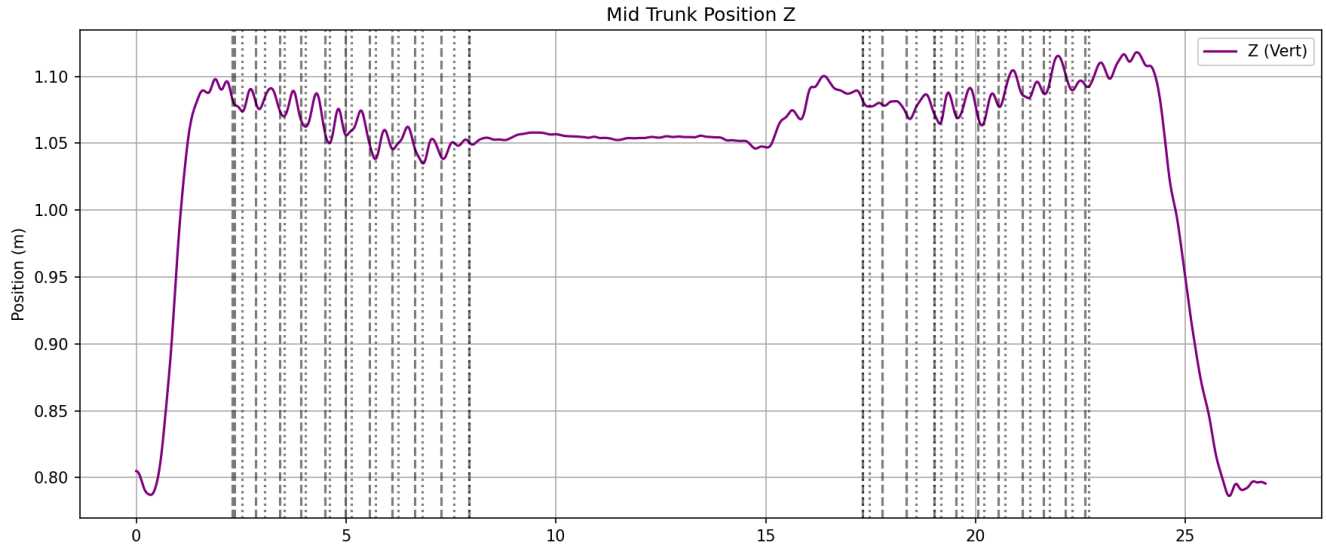

3D CoM Trajectory (Equal Aspect)
stand
gait forward
stop 5s
turn180
gait back
sit
Start
End
Z (Vert)
Y (AP)
X (ML)

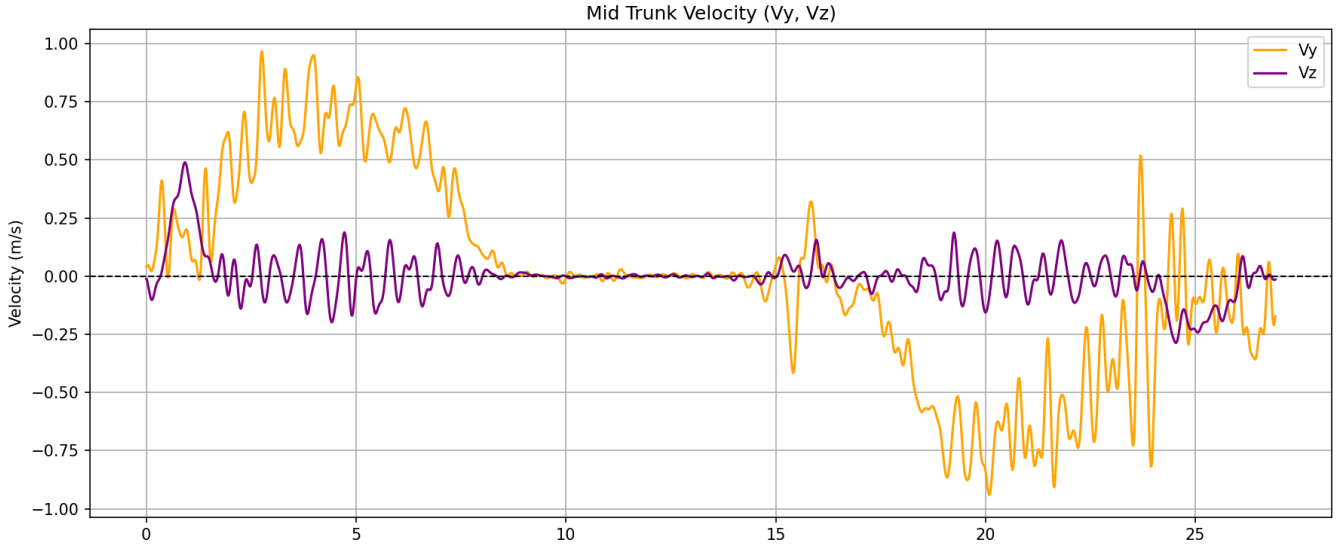

Sagittal View (YZ) - Progressive (Vy>0) vs Regressive (Vy<=0)
Z (Vertical) [m]
Y (Anteroposterior) [m]
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Right Path
Right Fwd (Vy > 0)
Right Bwd (Vy <= 0)
Right HS
Right TO
Left Path
Left Fwd (Vy > 0)
Left Bwd (Vy <= 0)
Left HS
Left TO

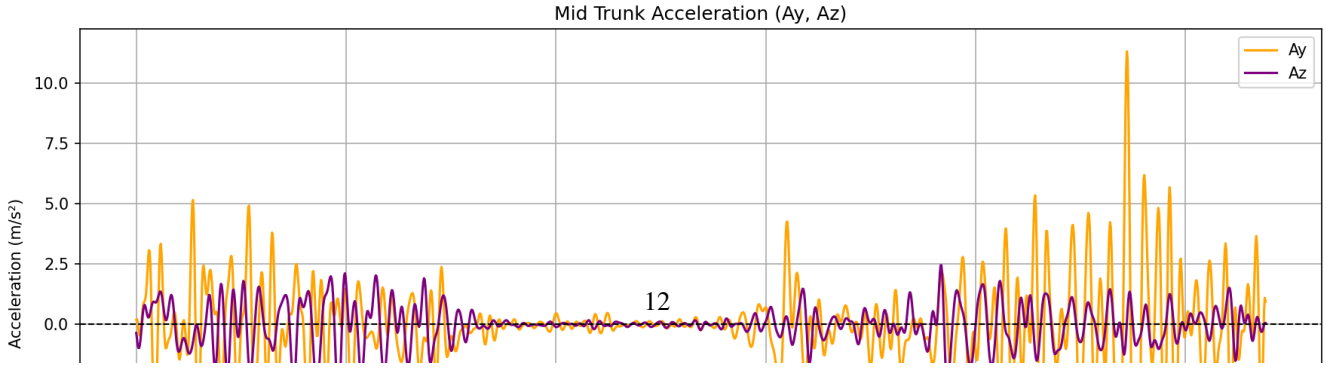

Right Mean Foot Position Y
Y (AP)
Position (m)
Left Mean Foot Position Y
Y (AP)
Position (m)
Right Mean Foot Position Z
Z (Vert)
Position (m)
Left Mean Foot Position Z
Z (Vert)
Position (m)
Right Mean Foot Velocity (Vy, Vz)
Vy
Vz
Velocity (m/s)
Left Mean Foot Velocity (Vy, Vz)
Vy
Vz
Velocity (m/s)
Right Mean Foot Acceleration (Ay, Az)
Ay
Az
Acceleration (m/s²)
Left Mean Foot Acceleration (Ay, Az)
Ay
Az
Acceleration (m/s²)
Right Mean Foot 2D Kinematics Magnitude (YZ)
|V_yz|
|A_yz|
Magnitude
Left Mean Foot 2D Kinematics Magnitude (YZ)
|V_yz|
|A_yz|
Magnitude

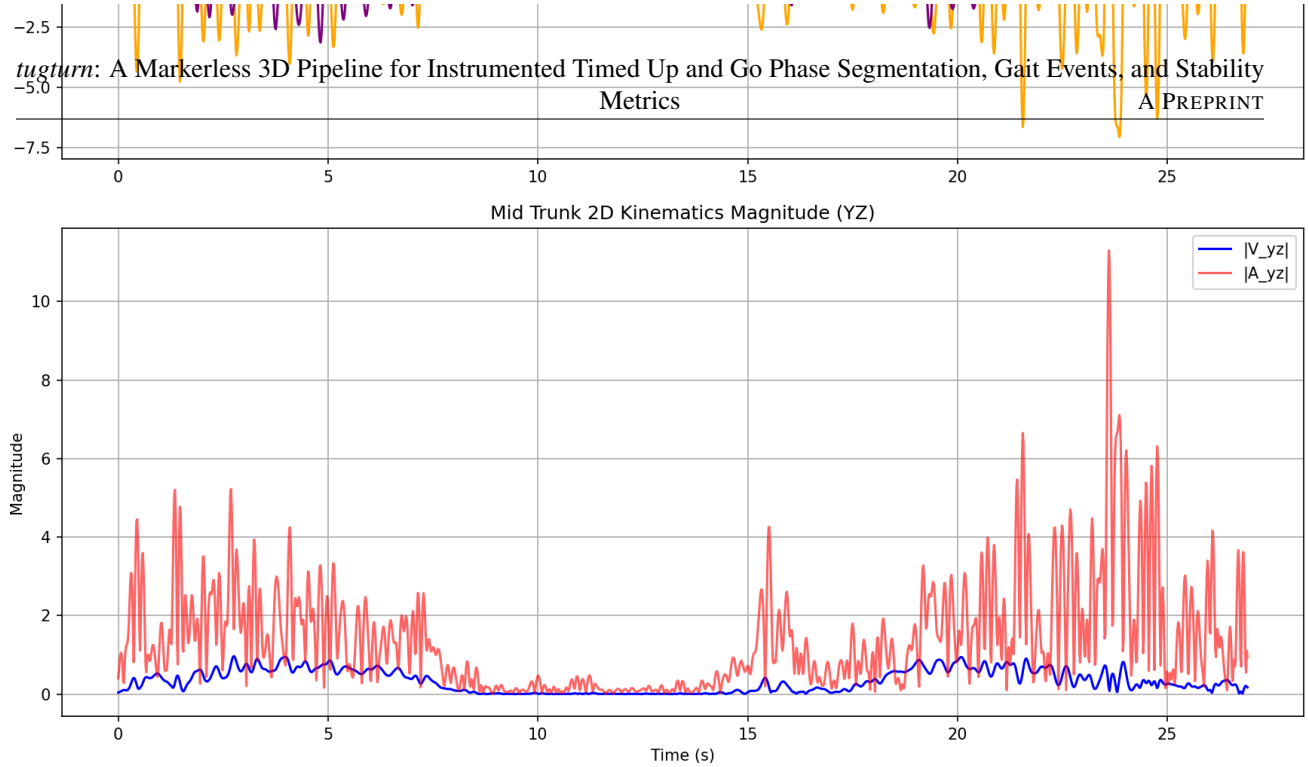

Time (s)
Time (s)
Mid Trunk Sagittal View (YZ) - Progressive (Vy>0) vs Regressive (Vy<=0)
Z (Vertical) [m]
Y (Anteroposterior) [m]
Mid Trunk Path
Fwd (Vy > 0)
Bwd (Vy <= 0)
Right HS (Trunk)
Right TO (Trunk)
Left HS (Trunk)
Left TO (Trunk)


Mid Trunk Position Y
Y (AP)
Position (m)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0
5
10
15
20
25


Mid Trunk Position Z
Z (Vert)
Position (m)
1.10
1.05
1.00
0.95
0.90
0.85
0.80
0
5
10
15
20
25


Mid Trunk Velocity (Vy, Vz)
Vy
Vz
Velocity (m/s)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25


Mid Trunk Acceleration (Ay, Az)
Ay
Az
Acceleration (m/s²)
10.0
7.5
5.0
2.5
0.0



Metadata

Field	Value
SEX	M
WEIGHT	87.3
HEIGHT	1.63
AGE	63.0
FPS	59.94005994005994
STAGE_HY	3.0
PERFORMANCE	4
TURNSIDE	R

Spatiotemporal Parameters

Global

Variable	Value
Cadence steps per min	67.912

Variable	Value
Velocity m s	0.347
XcoM Deviation Fwd m	0.027
XcoM Deviation Bwd m	0.041
Steps Walk Forward	12.000
Steps Walk Back	11.000
Fwd 1st strike inside zone	R
Fwd 1st complete step inside	L
Fwd Number of steps inside	11.000
Fwd Last strike inside zone	L
Fwd Sequence steps inside	['R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R', 'L']
Bck 1st strike inside zone	R
Bck 1st complete step inside	L
Bck Number of steps inside	10.000
Bck Last strike inside zone	R
Bck Sequence steps inside	['R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R']

Left vs Right

Variable	Left	Right
Stance Time s	0.691	0.694
Stance Time sd	0.104	0.029
Step Length m	0.321	0.368
Step Length sd	0.096	0.038
Step Width m	0.264	0.264
Step Width sd	0.077	0.063

Variable	Left	Right
Stride Length m	0.690	0.658
Stride Length sd	0.043	0.111
Swing Time s	0.409	0.365
Swing Time sd	0.040	0.043

Step-by-Step Timeseries (Gait Events / HS)

Step Index	Side	Time (s)	Y (m)	Phase	Stance Time (s)	Step Length (m)	Step Width (m)	Stride Length (m)	Sv Ti (s)
1	Right	2.286	1.155	gait_forward	0.2336	0.5956	0.1528	0.4829	0.:
2	Left	2.336	1.337	gait_forward	0.7174	0.1816	0.1575	0.686	0.:
3	Right	2.853	1.671	gait_forward	0.6673	0.3394	0.1905	0.7657	0.4
4	Left	3.420	2.034	gait_forward	0.6173	0.3877	0.2439	0.8206	0.4
5	Right	3.921	2.435	gait_forward	0.684	0.4186	0.2611	0.7642	0.:
6	Left	4.505	2.835	gait_forward	0.6173	0.4243	0.3158	0.6699	0.4
7	Right	4.988	3.187	gait_forward	0.7174	0.3612	0.2916	0.7095	0.:
8	Left	5.556	3.507	gait_forward	0.684	0.3382	0.2903	0.6746	0.4
9	Right	6.089	3.887	gait_forward	0.7174	0.3886	0.3068	0.5643	0.4
10	Left	6.640	4.178	gait_forward	0.9343	0.3031	0.2651	0.2732	0.:
11	Right	7.257	4.458	gait_forward	0.6673	0.2973	0.2315	0.0618	9.:
12	Left	7.925	4.456	gait_forward	9.376	0.1202	0.1066	0.3125	0.4
13	Right	17.301	4.409	gait_back	0.1668	0.138	0.1511	0.4787	0.8
14	Left	17.768	4.23	gait_back	0.8175	0.252	0.2024	0.6636	1.(
15	Right	18.352	3.901	gait_back	0.8175	0.3179	0.31	0.7745	0.:
16	Left	19.019	3.59	gait_back	0.6673	0.3882	0.3423	0.7806	0.:

Step Index	Side	Time (s)	Y (m)	Phase	Stance Time (s)	Step Length (m)	Step Width (m)	Stride Length (m)	Sw Ti (s)
17	Right	19.536	3.127	gait_back	0.6673	0.4157	0.3243	0.7066	0.3
18	Left	20.053	2.734	gait_back	0.6507	0.4221	0.3611	0.7134	0.4
19	Right	20.537	2.377	gait_back	0.7508	0.3749	0.3233	0.7374	0.3
20	Left	21.121	2.043	gait_back	0.634	0.4	0.3364	0.6386	0.3
21	Right	21.622	1.643	gait_back	0.684	0.3907	0.3236	0.5956	0.3
22	Left	22.139	1.411	gait_back	0.5672	0.3137	0.2791		
23	Right	22.606	1.148	gait_back		0.3742	0.3009		

TUG Phases (Seconds)

Phase	Range
stand	0.0s to 1.48s (Dur: 1.48s)
gait forward	2.17s to 7.97s (Dur: 5.81s)
stop 5s	7.97s to 14.51s (Dur: 6.54s)
turn180	14.51s to 17.3s (Dur: 2.79s)
gait back	17.3s to 23.06s (Dur: 5.76s)
sit	23.06s to 26.91s (Dur: 3.85s)
Total TUG Time	26.926900000000003
Turn Direction	Right

Figure 1: High-level *tugturn* processing flow used in this study.

References

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